we perform timing of inference speed for the main models of interest, on a TPUv3 accelerator; the difference between inference and backprop speed is a constant model-independent factor.

Figure 12 (left) shows how many images one core can handle per second, across various input sizes. Every single point refers to the peak performance measured across a wide range of batch-sizes. As can be seen, the theoretical bi-quadratic scaling of ViT with image size only barely starts happening for the largest models at the largest resolutions.

Another quantity of interest is the largest batch-size each model can fit onto a core, larger being better for scaling to large datasets. Figure 12 (right) shows this quantity for the same set of models. This shows that large ViT models have a clear advantage in terms of memory-efficiency over ResNet models.

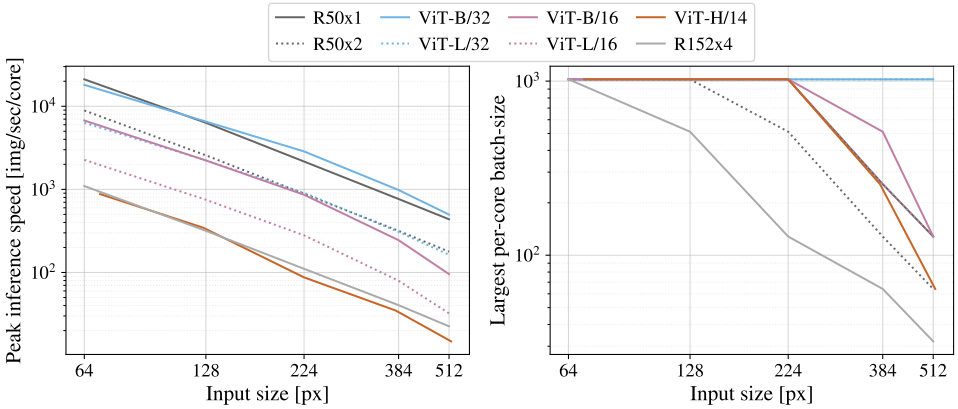

Figure 12: **Left:** Real wall-clock timings of various architectures across input sizes. ViT models have speed comparable to similar ResNets. **Right**: Largest per-core batch-size fitting on device with various architectures across input sizes. ViT models are clearly more memory-efficient.

D.6 Axial Attention

Axial Attention (Huang et al., 2020; Ho et al., 2019) is a simple, yet effective technique to run self-attention on large inputs that are organized as multidimensional tensors. The general idea of axial attention is to perform multiple attention operations, each along a single axis of the input tensor, instead of applying 1-dimensional attention to the flattened version of the input. In axial attention, each attention mixes information along a particular axis, while keeping information along the other axes independent. Along this line, Wang et al. (2020b) proposed the AxialResNet model in which all the convolutions with kernel size 3×3 in a ResNet50 are replaced by axial self-attention, i.e. a row and column attention, augmented by relative positional encoding. We have implemented AxialResNet as a baseline model.[3].

Moreover, we have modified ViT to process inputs in the 2-dimensional shape, instead of a 1-dimensional sequence of patches, and incorporate Axial Transformer blocks, in which instead of a self-attention followed by an MLP, we have a a row-self-attention plus an MLP followed by a column-self-attention plus an MLP.

Figure 13, present the performance of Axial ResNet, Axial-ViT-B/32 and Axial-ViT-B/16 on ImageNet 5shot linear, when pretrained on JFT dataset, verses the pretraining compute, both in terms of number of FLOPs and inference time (example per seconds). As we can see, both Axial-ViT-B/32 and Axial-ViT-B/16 do better than their ViT-B counterpart in terms of performance, but it comes at

[3]Our implementation is based on the open-sourced PyTorch implementation in `https://github.com/csrhddlam/axial-deeplab`. In our experiments, we reproduced the scores reported in (Wang et al., 2020b) in terms of accuracy, however, our implementation, similar to the open-source implementation, is very slow on TPUs. Therefore, we were not able to use it for extensive large-scale experiments. These may be unlocked by a carefully optimized implementation.

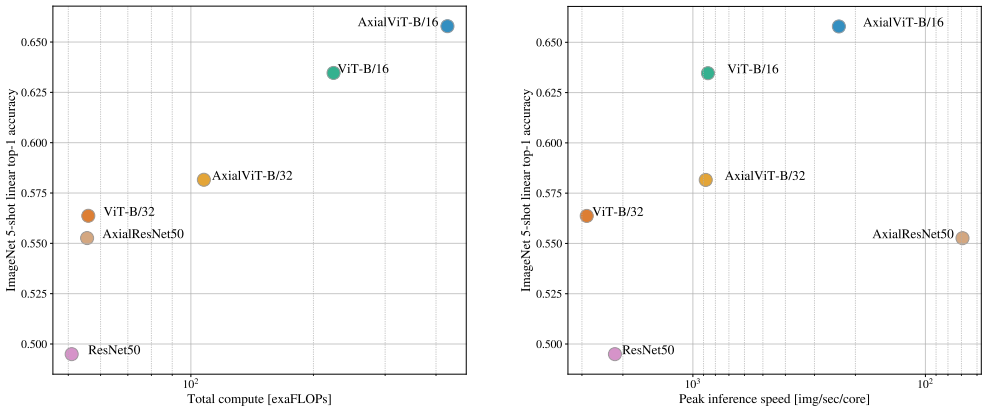

Figure 13: Performance of Axial-Attention based models, in terms of top-1 accuracy on ImageNet 5-shot linear, versus their speed in terms of number of FLOPs (**left**) and inference time (**left**).

the cost of more compute. This is because in Axial-ViT models, each Transformer block with global self-attention is replaced by two Axial Transformer blocks, one with row and one with column self-attention and although the sequence length that self-attention operates on is smaller in axial case, there is a extra MLP per Axial-ViT block. For the AxialResNet, although it looks reasonable in terms of accuracy/compute trade-off (Figure 13, left), the naive implementation is extremely slow on TPUs (Figure 13, right).

D.7 Attention Distance

To understand how ViT uses self-attention to integrate information across the image, we analyzed the average distance spanned by attention weights at different layers (Figure 11). This "attention distance" is analogous to receptive field size in CNNs. Average attention distance is highly variable across heads in lower layers, with some heads attending to much of the image, while others attend to small regions at or near the query location. As depth increases, attention distance increases for all heads. In the second half of the network, most heads attend widely across tokens.

D.8 Attention Maps

To compute maps of the attention from the output token to the input space (Figures 6 and 14), we used Attention Rollout (Abnar & Zuidema, 2020). Briefly, we averaged attention weights of ViT-L/16 across all heads and then recursively multiplied the weight matrices of all layers. This accounts for the mixing of attention across tokens through all layers.

D.9 ObjectNet Results

We also evaluate our flagship ViT-H/14 model on the ObjectNet benchmark following the evaluation setup in Kolesnikov et al. (2020), resulting in 82.1% top-5 accuracy and 61.7% top-1 accuracy.

D.10 VTAB Breakdown

Table 9 shows the scores attained on each of the VTAB-1k tasks.

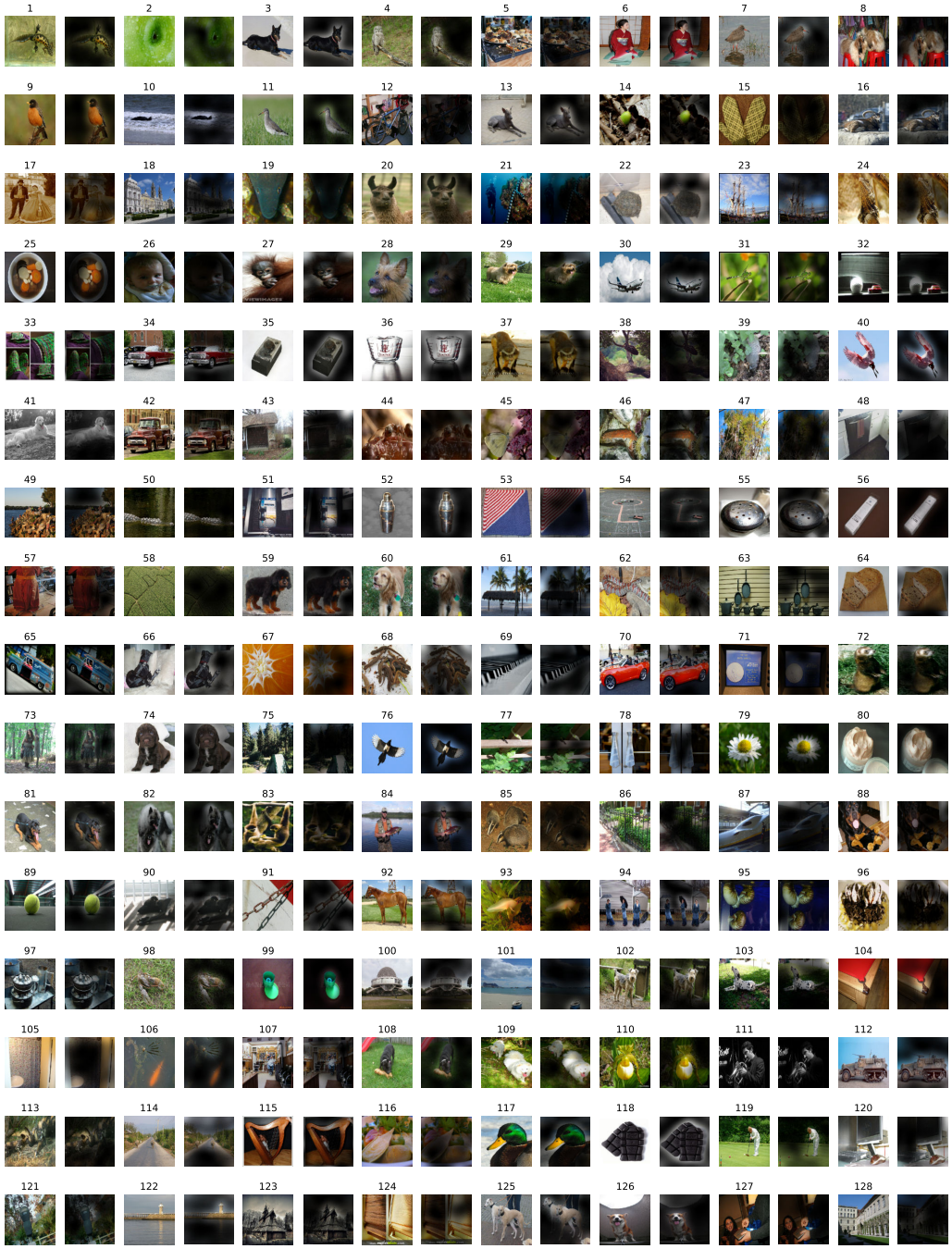

Figure 14: Further example attention maps as in Figure 6 (random selection).